Part D: Chatbot Report

1. Introduction

This report details the design, implementation, and features of the "Student Attendance Tracker," a conversational chatbot built in Python. The primary goal of this chatbot is to provide students with a simple, interactive, and efficient tool to monitor their class attendance. By automating the tracking process and providing instant feedback on attendance percentages and eligibility, the bot aims to help students stay on top of their academic requirements. The bot is designed to be user-friendly, persistent, and easily extendable.

2. Method and Implementation

The chatbot is implemented using a rule-based system in Python, leveraging regular expressions for intent recognition and JSON files for data management.

Core Components:

- **Input Handling & Intent Recognition**: The bot takes user input from the command line. It uses Python's re module to match keywords and patterns in the input string (e.g., \b(present|attended)\b) to determine the user's intent. The logic is structured as a series of if/elif conditions within the process_input method.
- **Data-Driven Design**: The chatbot's core knowledge, including subject names, icons, and response templates, is loaded from an external knowledge_base.json file. This approach decouples the bot's logic from its data, allowing for easy updates (like adding new subjects) without modifying the Python code.
- **State Management**: The bot maintains the state of the conversation through class attributes like self.student_name and self.current_subject. This allows it to remember who it's talking to and which subject is currently active for operations like marking attendance.
- **Data Persistence**: A key feature is the ability to save and load user data. The save_data function writes the current attendance records and student name to attendance_data.json upon exit. The load_data function reads this file on startup, ensuring that the user's progress is not lost between sessions.
- **Learning Component**: The bot includes a simple learning mechanism by remembering the user's name. After the name is provided once, it's saved and used in future greetings and messages for a personalized experience.

3. Input Cases and Outputs

Here are sample conversation flows demonstrating the chatbot's functionality:

Conversation 1: First-Time Use and Marking Attendance

- **Bot**: 🎓 Welcome to Student Attendance Tracker!... What's your name?
- **User**: my name is Sam
- **Bot**: Nice to meet you, Sam! 😊... 📚 Available Subjects: ... 1. ☁️ Cloud Computing...
- **User**: 1
- **Bot**: ☁️ Cloud Computing selected!... What would you like to do?
- **User**: present
- **Bot**: ✓ Marked PRESENT for Cloud Computing on 2025-10-18... ✨ What's next?

Conversation 2: Checking Percentage and Eligibility

- **User**: percentage
- **Bot**: 📊 Cloud Computing Attendance: ... • Percentage: 100.00% • Status: ✓ Eligible (Need 75%)
- **User**: Am I eligible?
- **Bot**: (This command is not implemented, would fall back) I'm not sure what you mean. Try... 'percentage' to check attendance.

Conversation 3: Checking Overall Summary and Finding Weakest Subject

- (After marking 'absent' for Big Data)
- **User**: all subjects summary
- **Bot**: 📚 Overall Attendance Summary: ✓ Cloud Computing: 100.00% (1/1) ⚠️ Big Data: 0.00% (0/1) ... 📈 Overall Average: 50.00%
- **User**: what is my lowest attendance
- **Bot**: ⚠️ Big Data has the lowest attendance: 0.00%... You need to attend the next 3 classes consecutively to reach 75%!

Conversation 4: Restarting and Loading Data

- (User quits and restarts the script)
- **Bot**: ✓ Knowledge base loaded... ✓ Previous attendance data loaded.
- **Bot**: 🎓 Welcome back, Sam! ... 📚 Available Subjects: ...

4. Features of the Chatbot

- **Dynamic Subject Management**: Subjects are loaded from knowledge_base.json, so new courses can be added without changing the Python code.
- **Persistent Memory**: The bot saves attendance data and the user's name, allowing sessions to be resumed later.
- **Comprehensive Attendance Tracking**: Users can mark themselves "present" or "absent" for any subject on a given day.
- **Detailed Analytics**:
    - Calculates attendance percentage for individual subjects.
    - Provides an overall attendance summary across all subjects.
    - Identifies the subject with the lowest attendance and advises on how to meet

eligibility criteria.

- **Eligibility Status**: Instantly tells the user if their attendance meets the required threshold (e.g., 75%).
- **Personalization**: Remembers and uses the student's name for a more engaging interaction.
- **Flexible Input**: Understands various ways of selecting subjects (by number or name) and triggering actions (e.g., 'percent', '%', 'status').
- **User-Friendly Menus**: Provides clear action menus to guide the user after each step.